

Scalability & Future Plan

Date	23 June 2026
Team ID	SWTID-2026-1377
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	1 Mark

Scalability & Future Plan:

This document describes how the **Voice-Based Concept Understanding Analyser** can be enhanced to support a larger number of users, expanded knowledge bases, and more advanced AI capabilities. It also outlines the future roadmap for improving system performance, accessibility, and learning effectiveness.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Application currently uses Flask for backend processing, which is not optimized for handling high concurrent user traffic.	Performance may degrade under heavy simultaneous usage.	High
2	Knowledge base contains predefined concepts and expected answers that require manual updates.	Limits scalability when supporting additional subjects and concepts.	Medium
3	No user authentication or learning history is maintained.	Users cannot track their learning progress or revisit previous analysis reports.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports moderate usage through the deployed Vercel application with Flask backend	Deploy backend using production servers (Gunicorn/uWSGI) with cloud infrastructure (AWS, Azure, or Render) and load balancing for improved scalability.
2	Data Storage	Uses JSON files as the knowledge base without persistent user data storage.	Integrate PostgreSQL or MongoDB to store user profiles, analysis history, and learning progress.
3	Performance	NLP processing performs efficiently for predefined concepts and moderate workloads.	Introduce caching, optimized semantic similarity algorithms, and asynchronous processing to improve throughput and response time.
4	Security	Basic input validation; no authentication or authorization mechanisms	Implement user authentication, secure APIs, input sanitization, rate limiting, and encrypted data storage for production deployment.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Production deployment using cloud infrastructure with scalable backend services and database integration	Q3 2026	Improved scalability, reliability, and support for a larger number of concurrent users.
Phase 3	Integrate advanced Generative AI models and Large Language Models (LLMs) for deeper semantic understanding, improved misconception detection, and richer personalized feedback.	Q4 2026	More accurate concept evaluation and higher-quality feedback for learners.
Phase 4	Add multilingual support, user authentication, learning progress tracking, dashboards, and mobile-friendly interface	Q1 2027	Enhanced accessibility, personalized learning experience, and broader adoption across educational institutions.